Decent Coaching Classes

VIII SB

Mind Your Math VIII

Time: 1 Hour **Marks: 30**

Q.1 (A) Four alternatives are given for each sub-question. Choose the correct one. (4)

(1) Which of the following is an irrational number?

(A) 22/7 (B) 3.14 (C) $\sqrt{2}$ (D) 0.3333…

(2) Two parallel lines are cut by a transversal. If one of the interior angles on the same side of the transversal is 65°, the other interior angle is

(A) 65° (B) 115° (C) 25° (D) 125°

(3) The power of $125^{2/3}$ is

(A) 5 (B) 3 (C) 2 (D) None of These

(4) The centroid of a triangle divides each median, measured from the vertex, in the ratio

(A) 2 : 1 (B) 1 : 2 (C) 1 : 1 (D) 3 : 1

Q.1 (B) Solve the following sub-questions. (4)

(1) Write "cube root of 17" in index form.

(2) Compare using < or > : $-5/6$ ____ $-7/8$

(3) A transversal intersects two lines at two distinct points. How many angles are formed in all?

(4) State whether the following statement is true or false: "The orthocentre of a right-angled triangle is the vertex of the right angle."

Q.2 Solve any four of the following sub-questions. (8)

(1) Expand: $(x + 5)(x - 3)$

(2) Factorise: $x^2 + 7x + 12$

(3) y varies directly as x. When x = 3, y = 12. Find the constant of variation and the value of y when x = 7.

(4) The diagonals of a rhombus are 24 cm and 10 cm. Find the length of its side.

(5) Line l || line m and line t is a transversal. The measures of a pair of corresponding angles are $(3x + 10)°$ and $(2x + 30)°$. Find the value of x.

(6) G is the centroid of Δ ABC and seg AD is a median. If l(AG) = 8 cm, find l(GD) and l(AD).

Q.3 Solve any two of the following sub-questions. (6)

(1) Expand: $(2m - 3)^3$

(2) 12 workers can complete a job in 15 days. How many days will 20 workers take to complete the same job?

(3) Mr. Desai sold a flat for ₹ 6,00,000 through an agent. The agent charged 2% brokerage from both the seller and the buyer. Find the total brokerage received by the agent and the amount received by the seller after paying the brokerage.

Q.4 Solve any two of the following sub-questions. (8)

(1) (a) Factorise: $2x^2 - 5x - 3$. (b) Factorise: $27p^3 + 8q^3$

(2) Construct quadrilateral ABCD such that l(AB) = 4.5 cm, l(BC) = 5 cm, l(CD) = 5.5 cm, l(DA) = 4 cm and l(AC) = 6.5 cm.

(3) A shopkeeper marks an article 20% above its cost price and then gives a discount of 10% on the marked price. Find his profit percent.